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Learning Multi-Modal Travel Speed from Remote Sensing and GPS Traces — Semester Plan

Target venue: IGARSS 2027, Reykjavík — **paper deadline 11 January 2027** (see §14)

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1. The question you're answering

In 1993 Waldo Tobler wrote down a hiking speed function that is still the default in GIS today:

```
walking speed (km/h) = 6 * exp( -3.5 * |slope + 0.05| )
```

Flat-ground speed of 5 km/h, a peak of 6 km/h on a gentle downhill, and two constants bolted on for other modes — multiply by 3/5 for off-path travel, by 5/4 for horseback. It was fit to figures from a Swiss cartography textbook.

Nearly every travel-time and accessibility model in geography still runs on it, or something like it. You're going to test it against real data, and then ask a harder question.

Question 1 — the measurement. What does the slope–speed relationship actually look like when you measure it from tens of thousands of real recorded hikes?

Question 2 — the one that makes this a remote sensing paper. OSM sometimes tells you a trail's `surface`, `sac_scale`, and `width`. Those tags are excellent predictors of speed — and they are *absent exactly where you'd want to predict.* So:

> **Can remotely sensed terrain and vegetation structure substitute for trail attributes that a map doesn't have?**

If overhead LiDAR and imagery can match what OSM tags buy you, then travel speed becomes predictable anywhere sub-meter data exists, with no map at all. That's the result worth having, and

it's what downstream lab work on off-road travel-time modeling needs — because that work has to condition on remotely sensed inputs, not on map attributes that don't exist off-trail.

Question 3 — the asymmetry. Tobler's function doesn't peak on the flat: it peaks at about $-2.9°$, a gentle downhill, and falls away in both directions from there. Uphill and downhill are not mirror images. Does real data agree, and where is the actual peak? See §8 for why this is cleanly measurable.

Question 4 — is mode a multiplier? This may be the strongest question in the list, and it depends on trace volume (Gate 1). Tobler handles other modes *multiplicatively* — $\times 3/5$ off-path, $\times 5/4$ on horseback — which assumes the slope–speed curve has the same shape for every mode and only its magnitude changes. OSM tells you which modes a trail permits (`foot`, `bicycle`, `horse`, `highway=bridleway|cycleway`, `mtb:scale`), so you can test that assumption directly. It is probably false for cycling: uphill riding collapses much faster than walking, and downhill is bounded by braking and technical difficulty rather than fitness. "Mode is a different curve, not a constant" would be a genuinely new result.

2. Why this problem is open

The data has existed for fifteen years and nobody has assembled it. The field splits into two camps that don't talk to each other. GIS and archaeology use Tobler because it's a closed form you can drop into a cost-distance raster. Deep travel-time estimation works on road networks with dense commercial traffic data and ignores terrain entirely.

The closest existing work is HikingTTE (2024), which learns hiking travel times with an LSTM and personalization — but from trajectories only, with no terrain covariates beyond slope, and no remote sensing at all.

Nobody has built the thing in the middle: **real observed speeds, real sub-meter terrain and vegetation structure, enough hikes to fit a conditional model, and an honest comparison against both the formula everyone uses and the map attributes everyone assumes they'll have.**

It also gets used the moment it exists, which is unusual — most first research projects produce nothing anyone touches for a year.

3. What you'll deliver

#	Deliverable	Why it matters
T1	**The trajectory dataset.** OSM GPS traces over one high-relief forested region — cleaned, stop-filtered, map-matched to the trail graph, with per-segment observed speed, direction of travel, and inferred travel mode (segments being fixed-length, see §8).	Your primary contribution. Nothing downstream can proceed without it.
T2	**The remote sensing covariate stack.** Terrain derivatives from 1 m LiDAR (slope, aspect, curvature, roughness), canopy cover, and vegetation structure from the point cloud where available.	A contribution in its own right, not plumbing — this is the input side of Question 2.
T3	**The four-arm comparison** (§4). Tobler vs. OSM tags vs. remote sensing vs. both, each with a heteroscedastic head predicting speed *and* variance.	The headline experiment. Question 2 lives here.

#	Deliverable	Why it matters
T4	**The empirical slope–speed curves** — directionally resolved, and per mode if volume allows.	The most citable figure in the paper.

T1 is the bulk of the work and it *is* a contribution. Six weeks of data engineering can feel like you haven't started doing research — you have. The modeling in T3 is fast once the data is clean, which is precisely why this is still open.

Core versus stretch. The core is T1, T2, and the Tobler comparison. That is a complete paper by itself, and it is the commitment. The following are all worth doing and worth keeping in view, but they are explicitly optional — dropping any of them costs you nothing:

- *Stretch* — per-mode curves and the multiplicativity test (Question 4), conditional on Gate 1 trace volume
- *Stretch* — per-hiker variance decomposition
- *Stretch* — a learned patch embedding in place of hand-crafted covariates (§9)
- *Stretch* — the "Both" arm of §4

Worth resisting the pull of a stretch until the core result is in hand — they're individually cheap and collectively how a semester overruns.

4. The four arms

This structure is the paper. Every arm predicts segment speed; they differ only in what they're allowed to see.

Arm	Conditioning	What it tells you
Tobler	slope only, fixed 1993 coefficients	The baseline everyone in the field knows
Tags	OSM trail attributes + slope	The "you already have a well-mapped trail" ceiling
Remote sensing	LiDAR + imagery covariates + slope, **no OSM tags**	**The contribution.** Can we predict speed without a map?
Both	everything	Is remote sensing complementary to tags, or redundant with them?

The interesting outcome is the Remote-sensing arm approaching the Tags arm. The interesting *negative* outcome is a large residual gap, which would tell you which trail properties matter for speed but aren't visible from above — also worth publishing, and more useful than a marginal win.

5. Week by week

This is a sketch to adapt, not a contract. The two gates and the week-13 draft are the parts worth holding to.

Weeks	What you're doing	Output
1–2	Region audit; trace reconnaissance; read the four core papers	Region comparison table
3	**GATE 1 — are the traces usable?** (§6)	Go / change-region decision

Weeks	What you're doing	Output
4–6	T1: bulk trace pull, cleaning, stop detection, map-matching	Trajectory dataset + data card
7–8	T2: terrain and vegetation covariate extraction, join to segments	Modeling table, one row per traversal
9	Tobler baseline; **the log-linear diagnostic** (§9)	Is log-speed linear in |slope + 0.05|?
10–11	T3: all four arms, heteroscedastic heads	**GATE 2** — does remote sensing close the gap?
12	T4: directional asymmetry on out-and-back pairs (§8)	Asymmetric slope–speed curves
13	**Full draft complete**	4-page paper draft
14	Figures, polish, per-hiker effects if they fit	Submission-ready draft
break	Revision and submission	**Submit by 11 January**

One thing to note: the draft lands in **week 13, not 14** — deliberately, because an unfinished draft at the wire can't be rescued over a holiday break.

If you slip, slip the back. **T1 + T2 + the Tobler comparison is a complete, publishable paper** even if nothing after week 11 lands.

6. The two gates

Gate 1 — Week 3: are the traces usable?

This can kill the project, so it goes first. OSM has four trace visibility levels and **they do not behave the same way**:

Visibility	What the API returns
identifiable	ordered points + timestamps, linked to user → **usable**
trackable	ordered points + timestamps, anonymous → **usable**
public	points **sorted by lat/lon**, ordering destroyed → a point cloud, not a track
private	nothing

Speed is uncomputable without temporal ordering. The question isn't "are there traces" — there are millions — it's **what fraction in your region are identifiable or trackable**.

- ☐ Pull 3–5 bounding boxes over the region via `/api/0.6/trackpoints?bbox=`
- ☐ Count returned `<trk>` elements; check which have monotonically increasing timestamps vs. scrambled coordinates
- ☐ For the usable ones, compute a naive speed histogram
- ☐ Apply crude stop-filtering (start with ~0.5 km/h; you'll tune it later) and recompute
- ☐ Break the usable count down by **permitted mode** — foot-only, bicycle-permitted, horse-permitted, mixed

Pass: enough ordered traces to be worth pursuing, and the filtered histogram sits in a plausible walking range (roughly 1.5–5 km/h) — with a faster second lobe if cyclists are present, which is a good sign.

Fail: almost everything comes back scrambled. Try another region before doing anything else, rather than starting to build.

The per-mode counts decide whether Question 4 is in scope. Bike and horse traces will be far rarer than walking — cyclists mostly upload to Strava and Trailforks, riders barely upload at all. If the volume is there, mode becomes a headline claim; if not, it costs nothing and becomes a limitation paragraph.

Write down the fraction. It goes in the paper, and nobody currently knows it.

Gate 2 — Week 11: does remote sensing close the gap?

- ☐ Every learned arm beats Tobler on held-out traces (MAE on segment speed)
- ☐ The Tags arm beats slope-only — confirms trail attributes matter at all
- ☐ The **Remote sensing** arm is reported against the Tags arm, with the gap quantified
- ☐ Predicted variances are calibrated, not just the means
- ☐ Your model beats a plain gradient-boosted-trees baseline on the same features (§9)

If nothing beats Tobler, that's a real and startling finding — a 1993 one-parameter exponential would be near the ceiling for on-trail hiking speed. Write it up honestly rather than hunting for a configuration that rescues the result.

7. Getting the data

Choosing a region

The criteria below follow from the research question, so they're firm. *Which* region you pick is your call — the audit in weeks 1–2 is what decides it, and the three candidates below are starting points for that audit, not a shortlist you have to choose from.

What the question requires:

- **Real relief.** Slope is the main effect; a flat region cannot answer Question 1.
- **A dense trail network**, with mode variety if Question 4 is to be in scope at all.
- **Forest cover**, or the vegetation-structure covariates have nothing to explain.
- **3DEP 1 m coverage** — and preferably a high-pulse-density acquisition, since pulse density strongly affects how well understory can be characterized (§7).
- **Enough identifiable or trackable traces** — this is Gate 1, and it's the one that can disqualify a region outright.

Candidate	Why it's on the list
White Mountains, NH	Dense AMC network, ~1400 m relief, heavy traffic, active mapping community
Cascades, WA	Greater relief, wider slope range
Colorado Front Range	Good relief and trace volume; less canopy, which cuts both ways

If the audit turns up somewhere better — more traces, better lidar, more mode variety — take it. Nothing downstream depends on which of these you pick.

Sources

Layer	Where
GPS traces	OSM `/api/0.6/trackpoints?bbox=` — max **0.25°** per bbox, 5,000 points per page, paginate with `page=`
Trail graph	Geofabrik extracts → `osmium` / `pyrosm` / `geopandas`
Elevation	`USGS/3DEP/1m` in Earth Engine; also OpenTopography and The National Map
LiDAR point clouds	OpenTopography — needed for vegetation structure, which the bare-earth DEM does not contain
Imagery	NAIP (1 m, 4-band RGB+NIR) via Earth Engine
Canopy	NLCD Tree Canopy Cover (30 m, CONUS) as the simple option
Land-cover embedding	AlphaEarth annual embeddings via Earth Engine — point-sampled, no raster work

The 0.25° cap means tiling — a region the size of the White Mountains is roughly 20–30 tiles. **Cache every response to disk on first fetch and never re-request.** You will re-run the pipeline dozens of times and should pay for the download once. Do not try the bulk GPX planet dump; the newest is from April 2013.

Verify 3DEP 1 m coverage against your exact footprint in week 1 — it's acquired project-by-project and is not seamless nationally.

Covariates worth computing

Terrain, from the 1 m DEM: slope, aspect, curvature, surface roughness, TPI. Use `gdaldem` and the Relief Visualization Toolbox.

Vegetation structure, from the point cloud: **normalized return density in understory height strata.** Read Campbell et al. (2018) before building this — they compared raw relative point density against a version normalized for how much pulse energy reaches the understory at all, and the normalized metric was far more predictive of measured understory density (R^2 0.44 against 0.14). They also model several thin strata (roughly 0.5–1.5, 1.5–2.5, 2.5–3.5 m) rather than one wide bin, and find pulse density the strongest positive influence on how well this works at all — which is why it's a region-selection criterion above.

Understory density is the thing that makes travel slow, and it's invisible in optical imagery and absent from a bare-earth DEM. If the Remote-sensing arm beats the Tags arm anywhere, this is the most likely reason. Point-cloud processing is also the slowest thing in T2, so extract it only within a buffer around traced trails — you don't need it where you have no observations.

Mode permissions, from OSM: `foot`, `bicycle`, `horse`, `highway` class, and `mtb:scale` / `sac_scale` where present. These do double duty — mode inference (§8) and difficulty covariates.

Season, from the trace timestamps: **day-of-year, sine/cosine encoded.** This is free and it matters more than it sounds — somewhere like the White Mountains, snow, ice, and mud season move hiking speed further than most terrain variables do. Unmodelled, seasonal variation lands in your residuals and inflates every variance estimate you report.

One extra column worth adding: alongside the hand-crafted covariates, sample **AlphaEarth's learned embedding** at each segment. It's point sampling in Earth Engine, so it costs almost nothing, and it buys you a comparison that's interesting on its own terms — does a general-purpose

learned representation match covariates designed specifically for this problem? It also makes the dataset directly usable by other lab work that conditions on those embeddings.

8. Handling the traces — where the real difficulty is

Read this section twice. It's the project.

Stop detection. Hikers stop constantly — views, lunch, photos, navigation, waiting. Raw point-to-point speed includes all of it, and if you don't remove stops you learn "hiking is slow" instead of anything about terrain. Pick a rule (speed threshold plus minimum duration), apply it consistently, and **report the fraction of time you dropped.** This one choice will move your headline numbers more than any model change.

Map-matching. Each trace point needs a trail segment so you can join covariates. Harder than road matching: the graph is sparse, trails run close together near junctions, and — the cruel part — **canopy degrades GPS accuracy exactly where the trails are.** Start with `leuvenmapmatching` or an HMM matcher (Newson & Krumm is the standard reference). Expect to hand-inspect failures and report a match rate.

Direction of travel. Timestamps give you direction along the matched segment, which Question 3 depends on. **Write a test for the sign convention** — an inverted uphill/downhill sign would silently invert your headline result and it is not obvious from the numbers.

Which mode is this trace? The trail network is not hiking-only, and OSM tells you so: `foot`, `bicycle`, and `horse` access tags, plus `highway=bridleway|cycleway|footway`, plus `mtb:scale` where someone has rated it for riding. A trace on a `bicycle=designated, foot=no` segment is a cyclist.

You have no explicit mode labels, so you infer mode from permitted modes plus observed speed. Done naively this is circular — inferring mode from speed, then modelling speed given mode. Anchor it instead: fit mode-conditional speed distributions **only on segments where permissions are exclusive**, then use those fitted distributions to classify traces on mixed-use segments. The modes separate well on speed alone (roughly 1.5–5 km/h walking, 6–13 on horseback, 10–25 cycling), so the anchoring is what keeps the argument honest rather than what makes it work. The mode-detection literature also finds *acceleration* about as discriminative as speed, which is worth trying if speed alone leaves the classes overlapping.

Report mode as inferred with a confidence, not as ground truth, and keep an unambiguous-permissions subset for validation.

Sampling rate varies wildly. Traces range from 1-second to 60-second intervals, and coarse sampling smooths over exactly the terrain variation you're measuring. Record the interval per trace; stratify or filter on it.

Out-and-back hikes are a gift. A hiker who goes up a trail and returns gives you an uphill and a downhill traversal of the *same segment*, by the *same person*, on the *same day* — same fitness, weather, and pack weight. That's a paired natural experiment for the asymmetry question that removes nearly every confound at once. Find these traces and treat them as a distinct, higher-quality subset.

Selection bias, which you must state and cannot fix. People who upload GPX traces to OSM are mappers and enthusiasts, not a random sample of hikers. Expect a skew toward fit, well-equipped, experienced walkers. Your model is a prior over *people who upload traces to OSM*. Say so plainly in the limitations.

Output schema — decide this in week 4

How reusable the dataset is gets fixed by a handful of choices made while you build it, and they're all expensive to change in week 12.

Resample to fixed-length segments rather than using OSM ways as they come. OSM ways run from a few meters to several kilometers, and a single mean speed over an 800 m way averages away exactly the slope variation you're trying to measure. Resample to constant along-path distance instead — 25 or 50 m are reasonable starting points — and pick the length by checking how much slope still varies *within* a segment at each candidate.

Store length and elapsed time per segment, not just speed. Speed is derived; keep what it was derived from. Anything that needs travel time rather than speed otherwise has to reconstruct lengths after the fact.

Make the direction convention an explicit column, not one that lives in your head. Signed slope along the direction of travel, written down, with the test from above.

Keep the cleaned point sequences, not only the aggregated table. The aggregate is one cut of the data at one resolution. Keeping the cleaned traces means it can be re-cut later without re-running the expensive parts.

One row per (trace × segment) traversal, carrying: segment id and geometry, length, elapsed time, speed, signed slope, direction, inferred mode and confidence, timestamp and day-of-year, contributor id, terrain and vegetation covariates, OSM tags, sampling interval, stop-filtered fraction, and map-match confidence.

9. Modeling approach

What follows is a proposal rather than a specification. If the data suggests something else once you have it, that's a better guide than this section.

What kind of problem this is

After T1 and T2 you'll have on the order of 10–30 tabular covariates and a scalar target. Two features of the data shape should inform the modeling more than anything else.

The effective sample size is the number of *traces*, not the number of segment-rows — points within one hike are heavily correlated, so millions of rows may represent only a few thousand independent units. Working that ratio out as soon as T1 lands tells you roughly how much model capacity the data can support.

And unlike most prediction problems, you already have a well-established parametric hypothesis for the dominant effect. That's an asset worth using rather than discarding.

The diagnostic that comes before any model

Tobler in log space is additive:

```
log v = log v0 - k * |s + s0|
```

so log-speed plotted against `|slope + 0.05|` should be a straight line of slope -3.5 if Tobler is right. This costs almost nothing and it tests the functional form itself rather than some model's error. A visibly curved relationship would be a more interesting finding than any accuracy improvement, and it would change what the paper is about — so it's worth doing in week 9, before fitting anything.

Proposed backbone: let Tobler's constants vary

Rather than learning speed from scratch, the suggestion is to keep Tobler's form and let its constants become functions of the covariates:

```
v = v0(x) * exp( -k(x) * |s + s0(x)| )
```

with peak speed `v0`, slope sensitivity `k`, and asymmetry offset `s0` each predicted from terrain, vegetation structure, and tags.

The appeal is less about accuracy than about what comes out. "Tobler's k = 3.5 is closer to 2.8 in open terrain and 4.6 under dense understory" is a result the field can absorb and argue with; a 12% error reduction from a black box is not. The learned parameters sit on the same scale as numbers people have used for thirty years. The form also extrapolates sensibly to steep slopes where observations will be thin, and `s0` is precisely the asymmetry Question 3 asks about — so the model's parameters and the paper's questions line up.

A natural extension is to allow different decay rates uphill and downhill instead of a single shared offset, which relaxes Tobler's particular assumption about the shape of the asymmetry.

The same move answers Question 4. Fitting `v0`, `k`, and `s0` **per mode** and comparing against a single shared curve with a fitted per-mode multiplier tests multiplicativity directly: if the multiplier version fits as well, Tobler's assumption holds; if the per-mode curves differ in `k` or `s0`, it doesn't. That comparison is cheap once the machinery exists, and it's the version of the result that transfers.

HikingTTE does something structurally similar — a modified-Lorentz slope-speed function embedded inside a neural model — so there's precedent for this over going end-to-end.

Output distribution

Speed is positive and right-skewed, and trace data carries real outliers: GPS glitches, and stops that slip through the filter. Both point toward working in log-space with a heavy-tailed likelihood rather than a Gaussian on raw speed. Log-space is also where Tobler is linear, so the two choices compose.

Hiker identity

Most hikers will contribute a single trace, so per-hiker terms need to pool toward the population mean rather than being estimated independently. Getting that right is what makes the week-14 variance decomposition meaningful rather than noise.

Baselines deserve real effort

On low-dimensional tabular data at this scale, gradient-boosted trees are a genuinely hard baseline — often harder than a neural network. Treating that as a serious comparison rather than a formality is the fastest way to learn whether the structured approach above is earning its complexity. Worth knowing that monotonicity in uphill grade is a constraint several implementations can enforce directly.

Bringing in remote sensing

Hand-crafted covariates — slope, roughness, understory return density — are the sensible starting point, and at this sample size they may well be the ending point too. A learned patch embedding in the RasterNet mould is the more ambitious version.

One caution worth planning around: an image encoder trained from scratch on a few thousand effective samples will overfit, and the symptom looks like ”remote sensing doesn't help” rather than ”we don't have enough data to tell.” A frozen pretrained encoder with a small trained head is the safer form of that experiment.

A cheap thing that might matter

Speed declines over a long hike. Cumulative ascent and elapsed time so far are two columns that capture much of that, and they're worth trying before reaching for anything that treats the trace as a sequence.

10. Metrics and evaluation

- **Primary:** MAE / RMSE on held-out segment speed, with Tobler as the reference baseline
- **Distributional:** NLL and CRPS for the heteroscedastic models — an unscored variance head is decoration
- **Calibration:** reliability plots and PIT histograms. A speed prior with badly calibrated uncertainty is unusable downstream
- **Stratified:** report everything binned by slope, and separately for the out-and-back subset

Split by trace and by geography, never by point. Points within one hike are massively correlated; a random point-level split leaks the same hiker's pace into train and test and hands you a beautiful, meaningless result. Hold out whole traces, and ideally whole trail systems.

11. Known hazards

Hazard	What to do
Traces come back with ordering stripped	Gate 1, week 3. Identifiable/trackable only
Stops contaminate speed	Explicit rule, consistently applied, fraction reported (§8)
Map-matching failures under canopy	Hand-inspect a sample; report match rate; drop low-confidence matches
Point-level splits leak	Split by trace and trail system (§10)
Sampling-rate heterogeneity	Record per trace; stratify or filter
Selection bias toward fit mappers	Cannot be fixed. State it
Season is a large unmodelled confound	Day-of-year is free from the timestamps. Encode it early (§7)
Bike and horse trace volume may be too thin	Counted at Gate 1. Question 4 is conditional on it, not assumed
Mode inference is circular if done naively	Anchor on exclusive-permission segments, then classify (§8)
Patchy 3DEP 1 m coverage	Verify your footprint in week 1
Point-cloud processing is slow	Vegetation structure needs the raw returns. Budget time, or subset to traced trails plus a buffer rather than the whole region
A learned image encoder overfits at this sample size	Freeze a pretrained encoder; train only a small head (§9)

12. Reading list

Grouped by when you'll need it rather than by importance. The first four in week 1; the rest when you reach the section they belong to.

Week 1 — the problem

1. **Tobler's hiking function** — the baseline you're testing. Short; read the actual formulation and note how thin its empirical basis is.
2. **HikingTTE** (2024) — closest prior work. LSTM, attention, a modified-Lorentz slope-speed function, personalized to individual walking ability. Trajectories only, no remote sensing. Understand precisely how you differ.
3. **Campbell et al.**, *Predicting the variability in pedestrian travel rates* (USDA Forest Service) — the most serious existing empirical measurement work, including off-trail rates.
4. **RasterNet** (arXiv 2006.08021) — the lab's approach to fusing LiDAR and overhead imagery in a geospatially consistent raster for a speed-regression target. Directly relevant to T2.

For the trace pipeline (§8)

5. **Newson & Krumm**, HMM map matching — the standard reference.
6. **An open GPS trajectory dataset and benchmark for travel mode detection** (arXiv 2109.08527) — what features people actually use for mode inference. Worth reading before you build the classifier: *acceleration* turns out to be about as discriminative as speed, which isn't obvious.

For the covariates (§7)

7. **Campbell et al. (2018)**, *Quantifying understory vegetation density using small-footprint airborne lidar*, Remote Sensing of Environment 215. Read this **before** computing understory metrics — it changes what you should compute. Note this is the same Campbell as #3: he works on both vegetation structure and pedestrian travel rates, which makes him worth following generally.

For modeling and evaluation (§9, §10)

8. **Gneiting & Raftery**, *Strictly Proper Scoring Rules, Prediction, and Estimation* (JASA, 2007) — the reference for CRPS and for what makes a probabilistic evaluation honest rather than decorative.
9. **Gelman & Hill** on multilevel models — you want the partial-pooling material for the per-hiker term, not the whole book.

Context, when you have time

10. **Hiking with Tobler: Tracking Movement and Calibrating a Cost Function** — calibrating Tobler from tracked movement.
11. *Improved prediction of hiking speeds using a data driven approach* (arXiv 2303.16065).
12. **Haklay (2010)**, *How good is volunteered geographical information?* — background for the OSM data-quality caveats you'll need in the limitations.

13. What the paper claims

Four pages holds roughly this much. Treat it as a target from week 1 and cut against it.

1. Here is a dataset of N map-matched traversals with observed speeds and 1 m terrain and vegetation covariates over *[region]* — how it was built, what fraction of OSM traces were usable,

and what's wrong with it.
2. Here is the slope–speed relationship measured from real hikes, and here is whether Tobler's exponential form holds.
3. Remotely sensed covariates recover X of what OSM trail tags buy you — so speed is predictable to within Y with no map at all.
4. The relationship is asymmetric by Z, measured on paired out-and-back traversals.
5. Mode is (or is not) a multiplier on a shared curve — if trace volume supports it.

Claims 1–3 are the paper. Claims 4 and 5 are the strongest figures if they land, and 5 is the one most likely to be cited. Everything else — per-hiker effects, the calibration analysis, the learned-embedding comparison — is supplementary material or the follow-on venue.

One precision note for the write-up: scope the claim explicitly to **on-trail, segment-level speed**. A model that predicts how fast someone moves along a mapped trail is not a general-purpose travel prior, and saying so in one sentence is cheaper than having a reviewer say it for you.

14. Venue and timeline

Primary: IGARSS 2027, Reykjavík, 11–16 July 2027.

Milestone	Date
Submission system opens	10 November 2026
Paper deadline	**11 January 2027**
Student Paper Competition deadline	11 January 2027
Acceptance notification	12 March 2027
Camera-ready	30 April 2027

Two implications you should design around from week 1.

The format is four pages, IEEE style. That is brutally short — it forces one headline result, one or two figures, and ruthless cuts. §13 is roughly what fits; everything in §3 will not. Decide by week 12 which single result leads: most likely the four-arm comparison, with the empirical slope–speed curve as the supporting figure. Write to the limit from the start rather than cutting a long draft down.

There is a Student Paper Competition with the same deadline. Worth entering.

Fallbacks and follow-ons, so a rejection isn't the end: **EARTHVISION** (CVPR workshop, deadlines around March) is a natural second shot and recoverable from a 12 March IGARSS notification. An extended version with the per-hiker analysis fits **SIGSPATIAL 2027** short papers (4 pages, deadline around June) or the **HuMob** workshop on human mobility prediction.

15. Working rhythm

- **Weekly meeting.** Bring what you did, what broke, and one plot. A plot that looks wrong is more useful than no plot.
- **Escalate at 2 days.** Stuck on the same thing for two days is a message, not a personal failing. Trace cleaning eats time in ways nobody predicts.
- **Keep a decisions log.** Every threshold, filter, and split boundary. You will need it for the data card and the paper's methods section, and you will not remember it in week 13.
- **Cache every API response on first fetch.** Download once.

- **Commit the pipeline, not just results.** The trace-cleaning code is a deliverable in its own right, and it's what makes the work survivable if someone else picks it up.